

Replication Package for: "The Hitchhiker's Guide to Markup Estimation: Assessing Estimates from Financial Data"

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This replication package accompanies De Ridder, Grassi and Morzenti (forthcoming). "The Hitchhiker's Guide to Markup Estimation: Assessing Estimates from Financial Data". Econometrica. DOI TBD.

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Instructions to replicators

The replication package is split into a `simulations` subfolder, which reproduces section 4 of the paper, and an `empirics` subfolder, which reproduces the empirical section 5 and section 6. All simulations can be reproduced on computers with the specifications described below. All empirical analysis must be performed in the CASD confidential French microdata environment.

The full replication of the simulations can be performed by running `launch.do` in the subfolder `simulations` of the replication package. Before running the file, change the path names at the top of that file to the local directory. All figures and tables will be reproduced in the subfolder `output`. The replicator should expect the code to run for 3 days. The replication of the simulation tables and figures uses Stata. The production of the simulated data uses Python and Matlab.

Any replication of the empirical analysis can only be performed by researchers with access to the confidential French micro data in the CASD environment. There are two sources of raw data. One is a panel of firms that was kindly provided by Burstein et al. (2025), the other is the EAP survey (Insee, 2009-2019). The EAP survey is a confidential micro dataset available through CASD. For details on how

to obtain permission to access confidential micro data in the CASD environment, see

<https://www.comite-du-secret.fr/>.

The raw panel from Burstein et al (2025), datafile ficusfare_profil_wof_19942019_v0222.dta, can be obtained through a transfer within the CASD environment. In order for a data transfer to be possible, the replicator must be authorized to access FARE (all years up to 2019), FICUS (all years), the community innovation survey (all years up to 2019), EAE: Services (all years), EAE Commerce (all years), EAE: Industry, EAE: BTP (all years). Researchers can contact the authors of Burstein et al. (2025) to request a file transfer within the CASD environment or, alternatively, replicators can contact Maarten De Ridder (m.c.de-ridder@lse.ac.uk) or Basile Grassi (basile.grassi@unibocconi.it) to arrange a file transfer within the CASD environment.

Once the raw datasets have been obtained, import the entire folder `empirics` from this replication package to the CASD environment using the standard procedure (imports-exports@casd.eu, see the CASD user manual). Paste the panel from Burstein et al. (2025) to the folder `rawdata`. In the main CASD data folder, find all the subfolders with EAP data. These have the format EAP_EAP_YYYY, where YYYY denotes the year. Paste all the folders EAP_EAP_2009 to EAP_EAP_2019 (so 11 folders in total) into the folder `rawdata` of the replication package.

You then need to change the path in 3 files.

- In the main directory of `empirics`, open the file `launch.do` and adjust the path to the directory where the folders in `empirics` are located.
- In the subfolder `datagen` of the main directory of `empirics`, open the file `master_datagen.do` and on line 10 adjust the path to the directory where the SAS file in `sasimport_april21` is located, which should be in the `datagen` folder.
- In the file `sasimport_april21`, which is placed in the subfolder `datagen`, replace the path `C:\Users\Public\Documents\DGM_ECMA` with the location of the `empirics` folder from the replication package. Leave the remainder of the pathname unchanged.

Then run `launch.do` with Stata. All figures and tables will be reproduced in the subfolder `output`.

Runtime is approximately 3 hours.

Computational requirements

Software requirements

- Stata (code must be run with version 17 or higher for the simulated data, version 14 or higher for the confidential data in the CASD environment). Additional packages are available through SSC (also in the CASD environment) and are added within the routine. An overview of additional files:

- `xtabond2` (as of 07-07-2025)
- `outreg2` (as of 07-07-2025)
- `winsor` (as of 07-07-2025)
- `xtreghdfe` (as of 07-07-2025)
- `binscatter` (as of 07-07-2025)
- `_gwtmean` (as of 07-07-2025)
- `md_ar1` (provided with the replication package)

Note: version 17 is needed for the simulations because python is called from stata within `simulations/launcher.do`. Replicators using older versions of stata can still reproduce the results by manually calling the python script in the replication package after the matlab simulation is complete. See the description of `launcher.do` for more details.

- Python 3.13. The following dependencies are installed (if needed) during the routine:
 - `pandas`
 - `numpy`
 - `pyreadstat`
- Matlab. The code was last run on version 2024B.
- The analysis of the confidential data furthermore requires an installation of SAS to convert the raw EAP SAS files to CSV files. SAS is available within the CASD environment.

Memory and runtime requirements

Simulations: The code was last run on a **8-core AMD Ryzen-based desktop (32gb ram, 4.7Ghz processors)**. Memory requirement: 32GB ram. Storage requirement: approximately 200GB. Computation took around 3 days. This includes generation of the simulated datasets. The code has only been run on windows and relies on a shell command in Stata.

Empirical analysis: The code was last run on the highest specification of the CASD server (Configuration 5) in July 2025. The highest specification is required because of the RAM required to produce the datasets (about 45GB). Storage requirement: approximately 10GB. Computation took 3 hours.

Data availability and provenance statements

Statement about rights

The authors of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Summary of availability

Some data **cannot be made** publicly available. Please see the section Instruction to Replicators above for details on how to obtain the confidential data.

Details on each data source

- (1) FICUS-FARE panel: We obtain the 1994–2019 raw panel with all balance sheet and income statement data from Burstein, et al. (2025). Access requires permission to use confidential data from the confidentiality committee (<https://www.comite-du-secret.fr/>) and a subscription to CASD (<https://www.casd.eu/en/>). In order for a data transfer to be possible, you must request access to FARE (all years up to 2019) and FICUS (all years), the community innovation survey (all years up to 2019), EAE: Services (all years), EAE Commerce (all years), EAE: Industry, EAE: BTP (all years). Access was obtained in summer 2020 and data was accessed between then and spring 2025. Burstein et al. (2025) have made their replication codes available in the central depository of the CASD environment. These cannot be exported from the environment, as they contain confidential firm identifiers. Please see the section *Instruction to Replicators* above for more details on how to obtain the panel.

Datafile (in the CASD environment DETVOLT):

C:/Users/Public/Documents/DGM_ECMA/rawdata/ficusfare_profil_wof_1994201_v0222.dta (not provided)

- (2) EAP data: We obtain data on quantities and prices for 2009 to 2019 from the Annual Production Survey (Enquete Annuelle de Production, EAP; Insee 2019). Access requires permission to use confidential data from the confidentiality committee (<https://www.comite-du-secret.fr/>) and a subscription to CASD (<https://www.casd.eu/en/>). You must request access to EAP (all years up to 2019). This will give a SAS file with the micro data with the name EAPYYYY.sas, where YYYY corresponds to the year of the survey. The replication code will fetch the raw SAS files and place them in the folder structure provided in the replication package. A data dictionary can be found in each of the micro data files' folder (the dictionary cannot be exported). Access was obtained in summer 2020 and data was accessed between then and spring 2025. We used the CASD project DETVOLT to access the data.

Datafiles (in the CASD environment):

Data/EAP_EAP_2009/eap2009.sas (not provided)

Data/EAP_EAP_2010/eap2010.sas (not provided)

Data/EAP_EAP_2011/eap2011.sas (not provided)

Data/EAP_EAP_2012/eap2012.sas (not provided)

Data/EAP_EAP_2013/eap2013.sas (not provided)

Data/EAP_EAP_2014/eap2014.sas (not provided)

Data/EAP_EAP_2015/eap2015.sas (not provided)

Data/EAP_EAP_2016/eap2016.sas (not provided)

Data/EAP_EAP_2017/eap2017.sas (not provided)

Data/EAP_EAP_2018/eap2018.sas (not provided)

Data/EAP_EAP_2019/eap2019.sas (not provided)

- (3) Deflators: The paper uses deflators for France. Sector deflators are obtained from EU-KLEMS (The Conference Board, 2018). The conference board website no longer links to the EU-KLEMS files. However, the files can be obtained from <https://dataverse.nl/dataset.xhtml?persistentId=doi:10.34894/6GDD70>. They are included in the replication package.

rawdata/gdp_deflator_euklems_nov21.dta (provided in the replication package)

rawdata/gdp_deflator_euklems_nov21.csv (provided in the replication package - non-prop. data format version)

Description of programs/code

High level overview of the file structure:

- Programs in `simulations/ datagen` generate the simulated data
- Programs in `simulations/ code` produce the tables/figures that generate the tables based on simulated data (Section 4 of the paper)
- Programs in `empirics/ datagen` produce the datasets for the empirical analysis (can only be run in the CASD environment)
- Programs in `empirics/ code` produce the tables/figures for the empirical analysis (can only be run in the CASD environment)

Program `launch.do`, in the `simulations` folder replicates the monte carlo analysis by calling `run_multi_file.m` and `MonteCarlo_producefilesforstata.py`, and then produces the

tables and figures by calling `ACF_estimator.do`, `DGM_estimator.do`, `table1.do`, `table2.do`, `table3.do`, `table11.do`, `figure1.do`, `figure2.do`, `figure3.do`

Description of programs in `simulation/ datagen`

- Program `run_multi_file.m` calls `main_baseline`, `main_alpha.m`, `main_eta.m`, `main_gamma.m`, `main_sigma.m`
- Program `main_baseline.m` produces the simulation of the main panel. It calls `RES_solve_sector_fixedfactor_prices_fixed.m`, `RES_solve_sector_translog_fixedfactor_prices_fixed.m`. It saves the panel in folder `export`.
- Program `RES_solve_sector_fixedfactor_prices_fixed.m` solves the model under cobb douglas, which is the first guess for the translog solution.
- Program `RES_solve_sector_translog_fixedfactor_prices_fixed.m` solves the full model with the translog production function.
- Program `main_alpha.m` produces the simulations with various values of the parameter alpha. It saves the panels in folder `export`.
- Program `main_eta.m` produces the simulations with various values of the parameter eta. It saves the panels in folder `export`.
- Program `main_gamma.m` produces the simulations with various values of the parameter gamma. It saves the panels in folder `export`.
- Program `main_sigma.m` produces the simulations with various values of the parameter sigma. It saves the panels in folder `export`.
- Program `MonteCarlo_producefilesforstata.py` takes the outputed panels in `export`, segments them in subsamples for the monte carlo exercise, and saves the files as .dta files in folder `output_montecarlo`.

Description of programs in `simulation/ code`

- Program `ACF_estimator.do` takes the simulated data and estimates production functions for revenue, storing the results in `analysisdata` (coefficients_byind_acf)
- Program `DGM_estimator.do` takes the simulated data and estimates production functions for quantity, storing the results in `analysisdata` (coefficients_byind_dgm)
- Program `table1.do` takes `coefficients_byind_acf` and `coefficients_byind_dgm`, merges these with the simulated data, and produces production function summary statistics in Table 1.
- Program `table2.do` takes `coefficients_byind_acf` and `coefficients_byind_dgm`, merges these with the simulated data, and produces markup summary statistics and correlates the estimates in Table 2.

- Program `table3.do` takes `coefficients_byind_acf` and `coefficients_byind_acf`, merges these with the simulated data, and regresses markup estimates and other simulated variables in Table 3.
- Program `table11.do` takes the simulated data, calculates correlations between prices and instruments, and produces Table 11.
- Program `figure1.do` takes `coefficients_byind_acf` and `coefficients_byind_acf`, merges these with the simulated data, and produces binscatters with the markup estimates for figure 1.
- program `figure2.do` adds different levels of measurement error to the output and revenue variables and then compares the performance of BB and ACF/DGM estimators. It then produces Figure 2a/b with the results.
- program `figure3.do` repeats the main markup estimations for panels with different calibrations of the production function parameters. For each calibration it calls `figure3_innerloop.do` which produces statistics on the markup estimates. It then plots the results in figure 3(a-h)
- program `figure3_innerloop.do` see description of `figure3.do`.
- program `tableF1.do` takes the simulated data and calculates correlations between instruments and prices, placing the results in table F1.

Program `empirics/launch.do`, if placed within the CASD DETVOLT environment, performs the data generation and table/results creation for all empirical analysis. It calls `master_datagen.do` and, `master_code.do`

Description of programs in `empirics/ datagen`

- Program `master_datagen.do` launches `sasimport_april21.sas`, `loadEAP.do`, `createdatasets.do`, `markups_estimation_bb.do`, `markups_estimation_dgmacf.do`, `markups_calculation.do`
- Program `sasimport_april21.sas` takes the raw EAP datasets from CASD and converts them into CSV files stored in rawdata.
- Program `loadEAP.do` loads the year-by-year EAP data, creates quantities and prices, and saves the firm-level panel `EAP_firmpanel` in `analysisdata`.
- Program `createdatasets.do` loads the FARE panel from Burstein et al. (2025), merges it with `EAP_firmpanel`, and performs various cleaning and deflating steps. Saves cleaned panel in `analysisdata`.
- Program `markups_estimation_bb.do` takes the panel from the preceeding step. Produces cleaned panel with production function variables called `smallestforpython`. Estimates Blundell-Bound production function and saves estimates in `analysisdata`.
- Program `markups_estimation_dgmacf.do` takes the cleaned panel and estimates the production function using the DGM (baseline) and ACF (revenue) methods and saves the estimates in `analysisdata`.

- Program `markups_calculation.do` takes the production function estimates and the cleaned panel and calculates all the markup series, saves them in `analysisdata`.

Description of programs in `empirics/ code`

- Program `master_code.do` launches `figure4.do` , `figure5.do` , `figureF1.do` , `table4.do` , `table5.do` , `table6.do` , `tableE1.do` , `tableE2.do` , `tableE3.do` , `tableG1G2.do`
- Program `figure4.do` takes the cleaned panel, merges with the markups, performs the standard outlier treatment, and plots binscatters for Figure 4 of the paper.
- Program `figure5.do` takes the cleaned panel, merges with the markups, performs the standard outlier treatment, calculates aggregate markups, and plots the aggregates for Figure 5 of the paper.
- Program `figureF1.do` takes the cleaned panel and calculates correlations between instruments and prices at the sector level. It then plots these for Figure F1.
- Program `table4.do` takes the cleaned panel, merges with the markups, produces summary statistics of markups for Table 4.
- Program `table5.do` takes the cleaned panel, merges with the markups, produces correlations between markup estimates for Table 5.
- Program `table6.do` takes the cleaned panel and merges it with the original panel for extra variables, merges with the markups, produces regressions of markups with other variables for Table 6
- Program `tableE1.do` takes the production function estimates and produces the average material output elasticities by sector for Table E1.
- Program `tableE2.do` takes the cleaned panel, merges with the single-stage markups, produces summary statistics of markups for Table E2.
- Program `tableE3.do` takes the cleaned panel, merges with the markups, produces correlations between markup estimates for Table E3.
- Program `tableG1G2.do` takes the cleaned panel and produces summary statistics (Table G1) and sector observation counts (Table G2).

License for Code

All code is licensed under a 3 clause (modified) BSD clause. See `LICENCE.txt` for details.

List of tables and figures

The provided code reproduces all numbers and figures in the paper. The table below identifies the file in which a table or figure is created. The sequence follows the sequence in which tables and figures are

included in the paper.

Figure/Table #	Program	Output file	Note
Table 1	simulations/code/table1.do	table1.xlsx	
Table 2	simulations/code/table2.do	table2.xlsx	
Table 3	simulations/code/table3.do	table3.xlsx	
Table 4	empirics/code/table4.do	table4.xlsx	Requires confidential data
Table 4	empirics/code/table4.do	table4.xlsx	Requires confidential data
Table 6	empirics/code/table6.do	table6q.tex,table6r.tex	
Figure 1	simulations/code/figure1.do	figure1a.pdf, figure1b.pdf	
Figure 2	simulations/code/figure2.do	figure2a.pdf, figure2b.pdf	
Figure 3	simulations/code/figure3.do	figure3a-h.pdf	
Figure 4	empirics/code/figure4.do	figure4a.pdf, figure4b.pdf	Requires confidential data
Figure 5	empirics/code/figure5.do	figure5a.pdf, figure5b.pdf	Requires confidential data
Table E1	simulations/code/tableE1.do	tableE1.xlsx	Requires confidential data
Table E2	simulations/code/tableE2.do	tableE2.xlsx	Requires confidential data
Table E3	simulations/code/tableE3.do	tableE3.xlsx	Requires confidential data
Figure F1	empirics/code/figureF1.do	FigureF1a, F1b, F1c.pdf	Requires confidential data
Table F1	simulations/code/tableF1.do	tableF1.do	
Table G1	empirics/code/tableG1G2.do	tableG1.xlsx	Requires confidential data
Table G2	empirics/code/tableG1G2.do	tableG2.xlsx	Requires confidential data

References

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